



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

MECHANICAL TECHNOLOGY: AUTOMOTIVE

2022

MARKING GUIDELINES

MARKS: 200

These marking guidelines consist of 16 pages.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS (GENERIC)

- | | | |
|-----|-----|------------|
| 1.1 | A ✓ | (1) |
| 1.2 | B ✓ | (1) |
| 1.3 | C ✓ | (1) |
| 1.4 | D ✓ | (1) |
| 1.5 | A ✓ | (1) |
| 1.6 | C ✓ | (1) |
| | | [6] |

QUESTION 2: SAFETY (GENERIC)**2.1 Rated speed of a grinding wheel:**

- Because the wheel could burst/break if it turns faster than its revolution range. / Avoid an accident. ✓
- Effectiveness of the grinding process will be compromised. ✓ **(Any 1 x 1)** (1)

2.2 Safety precautions of a band saw in operation:

- Never leave the band saw unattended. ✓
- Use a push stick when cutting. ✓
- Hold the work piece firmly and flat on the table. ✓
- Don't adjust the machine while working. ✓
- Don't open any guard while the machine is on. ✓
- Make relief cuts before cutting tight curves. ✓
- Don't force the material into the blade. ✓
- Keep hands clear from the action point. ✓
- Keep hands braced against the table. ✓
- Keep your hands on either sides of the blade and not in line with the cutting line and the blade. ✓
- Keep loose clothing clear from action point. ✓ **(Any 2 x 1)** (2)

2.3 Stages in which first aid is applied:

- Examination ✓
- Diagnosis ✓
- Treatment ✓ (3)

2.4 Causes of accidents:

- Unsafe acts ✓
- Unsafe conditions ✓ (2)

2.5 TWO advantages of the product layout:

- Handling of material is kept to a minimum. ✓
 - Time period of manufacturing cycle is less. ✓
 - Production control is almost automatic. ✓
 - Control over operations is easier. ✓
 - Greater use of unskilled labour is possible. ✓
 - Less total inspection is required. ✓
 - Less total floor space is needed per unit of production. ✓ **(Any 2 x 1)** (2)
- [10]**

QUESTION 3: MATERIALS (GENERIC)**3.1 Tempering:**

Tempering is a process generally applied to steel to relieve the strains/brittleness/improve ductility ✓ induced during the hardening process. ✓

(2)

3.2 Annealing:

- To relieve internal stresses ✓ that may have been set up during working of metal.
- To soften steel ✓ in order to facilitate the machining process.
- To refine their grain structure. ✓
- Reduce brittleness. ✓
- Make the steel ductile. ✓

(Any 3 x 1)

(3)

3.3 Normalising temperature:

- Above ✓ higher/upper critical temperature ✓
- Above ✓ AC₃ line. ✓

(Any 1 x 2)

(2)

3.4 Spark pattern for carbon steels:

3.4.1 High-carbon steel ✓

(1)

3.4.2 Low-carbon steel / Mild steel ✓

(1)

3.4.3 Cast-iron ✓

(1)

3.5 Carbon diagram:

- A. Temperature range / °C ✓
- B. AC₃ line / Higher/upper critical temperature line ✓
- C. AC₁ line / Lower critical temperature line ✓
- D. Carbon content / % carbon ✓

(4)

[14]

QUESTION 4: MULTIPLE-CHOICE QUESTIONS (SPECIFIC)

4.1	C ✓	(1)
4.2	B ✓	(1)
4.3	B ✓	(1)
4.4	D ✓	(1)
4.5	A ✓	(1)
4.6	C ✓	(1)
4.7	B ✓	(1)
4.8	A / C ✓	(1)
4.9	C ✓	(1)
4.10	B ✓	(1)
4.11	C ✓	(1)
4.12	A ✓	(1)
4.13	B ✓	(1)
4.14	D ✓	(1)
		[14]

QUESTION 5: TOOLS AND EQUIPMENT (SPECIFIC)**5.1 Wet compression test:**

- Pour/squirt some oil onto the piston through the spark plug hole. ✓
 - Do the compression test. ✓
 - Compare the readings to the dry compression test reading. ✓
- (3)

5.2 Functions:**5.2.1 Compression Test:**

Indicates ✓ the compression pressure ✓ created by piston in the cylinder. (2)

5.2.2 Cylinder Leakage Test:

- Indicates the % ✓ compressed air leakage from the cylinder. ✓
 - Indicates the location ✓ of the leakage from the cylinder. ✓
- (Any 1 x 2) (2)

5.3 Gas analyser:

- Inlet hose not to be stepped on. ✓
 - Hose connection must be airtight, and valve closed. ✓
 - No exhaust leaks. ✓
 - Condensate must be blown out of the hose and pick-up probe. ✓
 - Condenser must be drained after each test. ✓
 - Filter on the condenser stand must be changed regularly. ✓
 - Ensure test is done in a well-ventilated area. ✓
 - On a 12v analyser, the battery clamps must be cleaned. ✓
 - Make sure gas analyser is placed on a safe place. ✓
- (Any 4 x 1) (4)

5.4 Optical alignment gauges:

- Centre the steering on your car. ✓
 - Put each half of the tracking gauge against each of the front wheels. ✓
 - Look through the periscope and you should see / identify the alignment mark. ✓
 - Look through the periscope and align the vertical line with the triangle by moving the pointer arm. ✓
 - Read off the toe-in or toe-out. ✓
- (5)

5.5 OBD-II scanner:

- Plug the diagnostic tool into the OBD-II port. ✓
 - Enter the vehicle's details into the scanner. ✓
 - Turn on the vehicle's ignition. ✓
 - Start the diagnostic scan. ✓
 - Interpret the trouble codes and make a diagnosis. ✓
- (5)

5.6 Wheel balance methods:

- Dynamic balance ✓
 - Static balance ✓
- (2)

[23]

QUESTION 6: ENGINES (SPECIFIC)**6.1 Components driven by the crankshaft:**

- Flywheel ✓
- Camshaft ✓
- Supercharger ✓
- Connecting rod/Pistons ✓
- Oil pump ✓
- Water pump ✓
- Power steering pump ✓
- Air conditioning pump ✓
- Radiator fan ✓
- Distributor ✓
- Alternator ✓
- Transmission/gearbox ✓

(Any 4 x 1) (4)**6.2 Combustion engines:****6.2.1 Rotating mass:**

The crank pin, big-end ✓ and the lower two-thirds of the connecting rod. ✓

(2)**6.2.2 Reciprocating mass:**

The pistons, gudgeon pins ✓ and the upper third of the connecting rod. ✓

(2)**6.3 Advantages of a six-cylinder V-engine over a six-cylinder straight engine:**

- Can be mounted in smaller engine compartments. ✓
- Improved power to weight ratio. ✓
- More compact engine. ✓

(Any 2 x 1) (2)**6.4 Turbocharger:****6.4.1 Turbocharger parts:**

- A – Compressor/Compressor housing/casing ✓
- B – Exhaust gas out/discharge ✓
- C – Turbine wheel/blades ✓
- D – Exhaust gas in ✓
- E – Compressed air out/discharge ✓

(5)

6.4.2 Operation of the vanes in a variable geometry turbocharger at low speed:

- At low-speed range the variable nozzle vanes are almost closed. ✓
- The vanes create a narrow path to the exhaust turbine blades. ✓
- The angle of the vanes, directs the gases to hit the blades at the correct angle ✓
- This causes the turbocharger to spin faster. ✓

(4)

6.5 Types of superchargers:

- Roots ✓
- Twin-screw ✓
- Centrifugal ✓
- Eccentric / sliding-vane ✓

(4)

6.6 Disadvantages of superchargers compared to turbochargers:

- Superchargers are less effective at increasing engine power at high revolutions. ✓
- Superchargers use engine power to drive it (parasitic). ✓
- Higher fuel consumption if generated power is not fully used. ✓
- More space required to mount the Roots supercharger. ✓
- Roots and twin-screw superchargers deliver air in bursts. ✓
- It is more expensive than a turbocharger. ✓

(Any 3 x 1)

(3)

6.7 Difference between twin-turbocharging and twin-charging:

Twin-turbocharging uses two turbochargers ✓ while twin-charging uses a combination of a turbocharger and a supercharger. ✓

(2)

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QUESTION 7: FORCES (SPECIFIC)**7.1 Definitions:****7.1.1 Work:**

Work is done when a force ✓ overcomes resistance and causes movement. ✓ (2)

7.1.2 Clearance volume:

This is the volume above the crown of the piston, ✓ when the piston is at TDC / combustion chamber. ✓ (2)

7.2 The mean effective pressure represented:

7.2.1 Indicator diagram / Pressure-volume diagram ✓ (1)

7.2.2 kPa/Pa or kN/m² /Nm² ✓ (1)

7.3 Calculations:**7.3.1 Swept Volume:**

$$\begin{aligned}SV &= \frac{\pi D^2}{4} \times L \\&= \frac{\pi \times 7^2}{4} \times \frac{65}{10} \checkmark \\&= 250,15 \text{ cm}^3 \checkmark\end{aligned} \quad (3)$$

7.3.2 Original clearance volume:

$$\begin{aligned}CV &= \frac{SV}{CR - 1} \\&= \frac{250,15}{9 - 1} \checkmark \\&= 31,27 \text{ cm}^3 \checkmark\end{aligned} \quad (3)$$

7.3.3 Stroke length:

$$\begin{aligned}
 SV &= CV (CR - 1) \quad \checkmark \\
 &= 31,27(10 - 1) \quad \checkmark \\
 &= 281,42 \text{ cm}^3 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 SV &= \frac{\pi D^2}{4} \times L \\
 L &= \frac{SV \times 4}{\pi \times D^2} \quad \checkmark \\
 &= \frac{281,42 \times 4}{\pi \times 7,2^2} \quad \checkmark \\
 &= 6,912 \text{ cm} \quad \checkmark \\
 &= 69,12 \text{ mm} \quad \checkmark
 \end{aligned}$$

(7)

7.4 Calculations:**7.4.1 Indicated power:**

$$\begin{aligned}
 L &= \frac{10}{100} \\
 &= 0,1 \text{ m} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{Area} &= \frac{\pi D^2}{4} \\
 &= \frac{\pi \times 0,08^2}{4} \quad \checkmark \\
 &= 5,03 \times 10^{-3} \text{ m}^2 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 N &= \frac{2500}{60 \times 2} \quad \checkmark \\
 &= 20,83 \text{ firing strokes/sec} \quad \checkmark
 \end{aligned}$$

$$IP = PLANn$$

$$\begin{aligned}
 IP &= (1250 \times 10^3) \times 0,1 \times (5,03 \times 10^{-3}) \times 20,83 \times 4 \quad \checkmark \\
 &= 52387,45 \text{ W}
 \end{aligned}$$

$$IP = 52,39 \text{ kW} \quad \checkmark$$

(7)

7.4.2 Torque:

$$BP = 2\pi NT$$

$$N = \frac{2500}{60}$$
$$= 41,67 \text{ r/s} \quad \checkmark$$

$$T = \frac{BP}{2\pi N} \quad \checkmark$$
$$= \frac{(46,08 \times 10^3)}{2 \times \pi \times 41,67} \quad \checkmark$$
$$= 176 \text{ Nm} \quad \checkmark$$

(4)

7.4.3 Mechanical efficiency:

$$\text{Mechanical efficiency} = \frac{BP}{IP} \times 100$$
$$= \frac{46,08}{52,39} \times 100 \quad \checkmark$$
$$= 87,96 \% \quad \checkmark$$

(2)
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QUESTION 8: MAINTENANCE (SPECIFIC)**8.1 Lean fuel mixture:**

- High oxygen (O₂) ✓
- Low Carbon dioxide (CO₂) ✓
- High Nitrogen oxide (NO_x) ✓

(3)

8.2 High hydrocarbon (HC) exhaust gas reading:

- Incomplete combustion ✓
- Improper valve timing ✓
- Improper ignition timing ✓
- Faulty air management system ✓
- Blocked or restricted air-filter ✓
- Faulty temperature sensor ✓
- Faulty oxygen sensor ✓
- Excessive fuel pressure ✓
- Non-functioning PCV valve ✓
- Faulty catalytic convertor ✓

(Any 3 x 1)

(3)

8.3 Compression test:

Causes	Corrective Measures
• Blown cylinder head gasket ✓ • Cracked cylinder head ✓	• Replace with new gasket ✓ • Replace/repair cylinder head ✓

(4)

8.4 Bubbles in the radiator water:

- Blown cylinder head gasket ✓
- Cracked cylinder head ✓

(2)

8.5 Oil pressure test:

- Oil pressure at idling speed. ✓
- Oil pressure at high revolutions. ✓
- Oil pressure when engine is cold. ✓
- Oil pressure when engine is hot. ✓

(Any 3 x 1)

(3)

8.6 Precautions while setting up the fuel tester:

- Ensure that you wear the correct PPE e.g. safety goggles. ✓
- Ensure that the tester can read the fuel pressure of the engine. ✓
- Ensure that the place where you will couple the tester is clean before you remove the sender unit. ✓
- Ensure the rubber pipe on the tester is not perished. ✓
- Put the tester at a place on the engine that is safe from the running engine. ✓
- Ensure that there are no flammable materials in proximity. ✓
- Ensure that there is a fire-extinguisher. ✓

(Any 4 x 1)

(4)

8.7 Radiator pressure drop:

- Repair leaks between components / gasket leaks. ✓
- Repair leaking hoses. ✓
- Tighten loose hose clamps. ✓
- Repair or replace leaking water pump. ✓
- Repair or replace corroded pipes. ✓
- Replace blown head gasket. ✓
- Repair or replace leaking radiator. ✓
- Repair or replace leaking cabin heater radiator. ✓
- Renew the heater tap. ✓
- Renew the welch or core plugs. ✓

(Any 4 x 1)**(4)
[23]**

QUESTION 9: SYSTEMS AND CONTROL (AUTOMATIC GEARBOX) (SPECIFIC)**9.1 Lock-up torque converter:****9.1.1 Function of the lock-up clutch:**

- The lock-up clutch improves efficiency ✓
- Prevents slip ✓

(2)

9.1.2 Functions of a stator:

- Redirects oil back to the impeller. ✓
- Increases the engine torque. ✓

(2)

9.1.3 Lock-up clutch is engaged:

- The oil pressure in the torque converter increases with engine speed. ✓
- The pressurised oil is channelled to the lock-up clutch piston. ✓
- The lock-up clutch piston pushes the friction plate against the clutch friction surface attached to the housing. ✓
- Since the friction plate is splined to the impeller, it connects the impeller and turbine. ✓
- The turbine and impeller begin to turn as one. ✓

(5)

9.2 Automatic gearbox:**9.2.1 Double epicyclic gear train. ✓**

(1)

- Three forward ✓
- One reverse ✓

(2)

9.3 Cooling the hydraulic transmission fluid in an automatic transmission:

- The hydraulic transmission fluid is circulated through an oil cooler at the radiator. /Oil is cooled by circulating through the bottom tank of the radiator. ✓
- The airflow over the transmission sump allow for cooling of the oil. ✓

(2)

9.4 Differences between the construction of a manual transmission and automatic transmission:

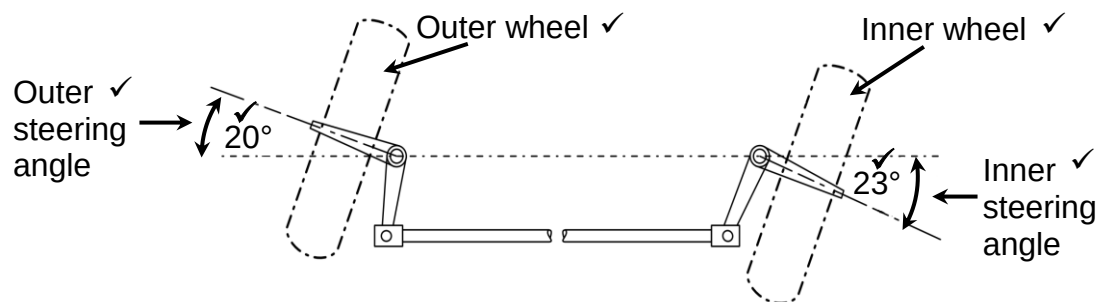
Manual	Automatic
• Less complex design. ✓ • Different gear sets used to obtain different gear ratios. ✓ • Cluster and simple gear trains used. ✓ • Dry clutch used when changing gears. ✓	• More complex design. ✓ • Same gear sets used to obtain different gear ratios. ✓ • Epicyclic-gear trains used. ✓ • Wet clutch used to engage gears. ✓

(Any 2 x 2)

(4)
[18]

QUESTION 10: SYSTEMS AND CONTROL (AXLES, STEERING GEOMETRY AND ELECTRONIC) (SPECIFIC)**10.1 Pre-checks on vehicle suspension before wheel alignment:**

- Correct preload on the wheel (hub) bearings ✓
- Kingpins and bushes ✓
- Suspension ball joints for wear, locking and lifting ✓
- Suspension bushes for excessive free movement ✓
- Steering box play and whether secure on chassis ✓
- Tie-rod ends ✓
- Sagged springs, which includes riding height ✓
- Ineffective shock absorbers ✓
- Spring U-bolts ✓
- Chassis for possible cracks and loose cross-members ✓

(Any 3 x 1) (3)**10.2 Toe out on turns:**

NOTE: Steering angles should be different. If degrees indicated are the same, candidate loses the TWO marks. Angles sizes shown are just an example.

(6)**10.3 Faults toe-out on turns:**

- Wear on the suspension parts ✓
- Wheel bearing wear ✓
- Steering system wear ✓

(Any 2 x 1) (2)**10.4 Static balancing:**

- Mount the wheel so that it is free to spin on a spindle. ✓
- Spin the wheel slowly. ✓
- If the wheel is out of balance, it will always come to rest at the same point, ✓ the 'heavy spot', at the bottom.
- To correct static imbalance, a small weight is fitted to the wheel rim by trial and error, opposite the 'heavy spot'. ✓
- Repeat until the wheel stops at random positions. ✓

(5)

10.5 Electronic Control Unit (ECU) functions:**10.5.1 Air-induction system:**

The air-induction system measures ✓ and controls ✓ the air required for the combustion. (2)

10.5.2 Ignition system:

The purpose of the ignition system is to ignite ✓ the air/fuel mixture in the combustion chamber at the correct time. ✓ (2)

10.6 Catalytic convertor gases:

- Hydrocarbons (HC) ✓
 - Carbon monoxide (CO) ✓
 - Nitrogen oxide (NOx) ✓
- (Any 2 x 1) (2)

10.7 Labels common rail direct injection (CRDI) system:

- A. Common rail ✓
 - B. High pressure pump / pump / diesel pump ✓
 - C. Diesel/fuel filter ✓
 - D. Injectors ✓
- (4)

10.8 Function of the pressure regulator:

- It keeps the pressure ✓ in the common rail at a specified pressure. ✓
 - It relieves excessive pressure ✓ in the common rail. ✓
- (Any 1 x 2) (2)

10.9 The alternator:**10.9.1 Component:**

- A. Rotor ✓
 - B. Capacitor ✓
- (2)

10.9.2 Winding connection:

Star / Y ✓ connected stator windings (1)

10.9.3 Diodes:

Six (6) ✓ (1)
[32]

TOTAL: 200